

QRM II Graded Assignment (3)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie revenue’.
- a. Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. **[2 points]**

- b. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. **[2 points]**
- d. Next, estimate the model

$$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. **[2 points]**

- e. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
- g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

step a

```
#model genaamd "model4.1" is een lineair model, hoe beweegt omzet met gegeven budget.
model4.1 <- lm(revenue ~ budget, data = movies)
summary(model4.1)
```

```
##
## Call:
## lm(formula = revenue ~ budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -387813422  -39322110   610960   8901755  950609114
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.307e+06  4.636e+06  -0.713    0.476
## budget      2.830e+00  8.798e-02  32.169   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 109900000 on 907 degrees of freedom
## Multiple R-squared:  0.5329, Adjusted R-squared:  0.5324
## F-statistic: 1035 on 1 and 907 DF, p-value: < 2.2e-16
```

Your Answer:

The formula used, $\text{Revenue}_i = 0 + 1\text{Budget}_i + \epsilon_i$, so does the revenue move proportionally with the added budget, so is it profitable?

step b

```
#wat is de richtingscoefficient van het lineaire model(regressie model)
coef(model4.1)
```

```
## (Intercept)      budget
## -3.307163e+06  2.830122e+00
```

Your Answer:

The answer gotten above is $b_1 = 2.83$, meaning every dollar spent on the budget on average, results in 2.83 revenue.

step c

```
#geestimeerde helling b1 en SE vinden, na het vinden vna b1, heeft R 2 punten genomen,
↪ hoeveel blijven er over om te ondervinden of het klopt, test hoeveel SE meer dan 1
↪ zijn
b1 <- coef(summary(model4.1))["budget", "Estimate"]
se1 <- coef(summary(model4.1))["budget", "Std. Error"]
df1 <- df.residual(model4.1)
t_stat <- (b1 - 1) / se1
p_value <- 1 - pt(t_stat, df1)
print(p_value)/print(b1)
```

```
## [1] 0
## [1] 2.830122
```

```
## [1] 0
```

Your Answer:

$b_1 = 2.83$, $se_1 = 0.088$, so $t_stat = 20.8$, and the p-value is basically 0. This means the probability of observing a slope as extreme as 2.83, assuming the true slope is 1 or less, is practically zero, so we reject the null hypothesis and conclude that Chantal is right for this dataset.

step d

```
#een log = 0 kan niet, daarom die eruit filteren
movies$log_revenue <- ifelse(movies$revenue == 0, NA, log(movies$revenue))
movies$log_budget <- ifelse(movies$budget == 0, NA, log(movies$budget))
model4.2 <- lm(log_revenue ~ log_budget, data = movies)
summary(model4.2)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget, data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.8586  -0.6783   0.1831   0.8499   6.6344
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.56490     0.73689   3.481 0.000535 ***
## log_budget   0.88520     0.04311  20.535 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.479 on 627 degrees of freedom
## (280 observations deleted due to missingness)
## Multiple R-squared:  0.4021, Adjusted R-squared:  0.4012
## F-statistic: 421.7 on 1 and 627 DF, p-value: < 2.2e-16
```

Your Answer:

Movies with a revenue or budget of 0, are called NA before taking logs, because $\log(0)$ is NA, 629 movies have both a positive revenue and budget.

step e

```
#wat is de rc van LM2
coef(model4.2)
```

```
## (Intercept) log_budget
##  2.5648963    0.8852038
```

Your Answer:

$b_1 = 0.885$, meaning, a 1% increase in budget, results in a 0.885% increase in revenue, so there are signs of diminishing returns, as a log model with an elasticity below 1 is inelastic.

step f

```
#using the anova formula (what google said) isnt optimal, because the variables are not
↪ the same. So the other easier option, using R2
summary(model4.1)$r.squared
```

```
## [1] 0.5329176
```

```
summary(model4.2)$r.squared
```

```
## [1] 0.4021191
```

Your Answer:

model4.1 has a R^2 of 0.533 and model4.2 a R^2 of 0.402, going off of this, model4.1 is better. However, this method is not bullet proof, because revenue is compared with $\log(\text{revenue})$, and model4.1 has 909 movies, and model4.2 has 629. Both models take very large movies into account, however because model4.1 uses revenue and budget in dollars and model4.2 uses percentages, the large movies do not have so much impact in model4.2. So, taking everything into account model4.2 gives a more realistic outcome, that is the better fit.

step g

#not needed

Your Answer:

Only looking at model4.1, chantal would have been right in my eyes, but with the additional information of model4.2. I think Bob is more right, but not fully, but because of the budget's diminishing returns it stops being reasonable to invest somewhere. So my advice to the studio, be somewhere in the middle, do not take such extreme positions. If you have the money to invest in the budget, it is most likely profitable to do so.

2

- Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. [3 points]
- Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. [3 points]

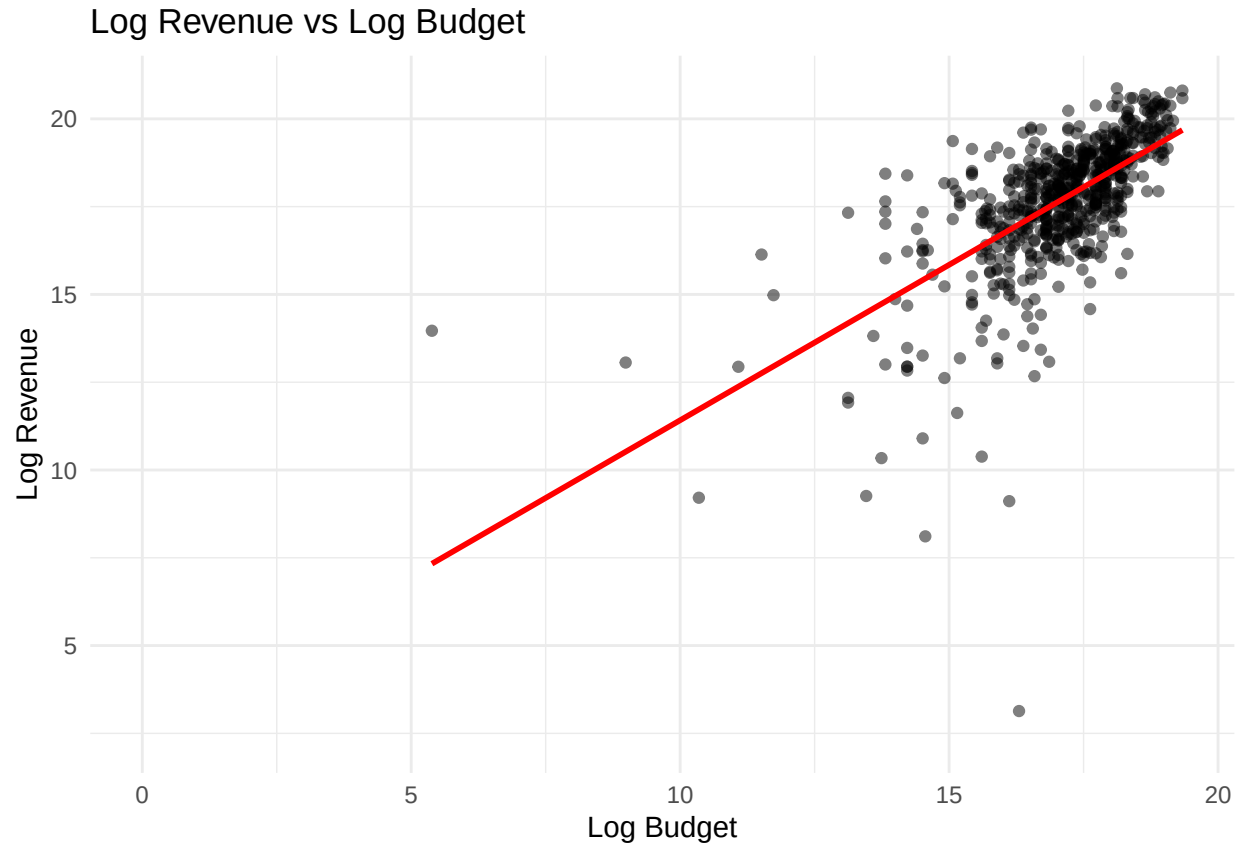
step a

```
#Alpha = 0.5 omdat je het dan laat doorschijnen, zo zie je wanneer films dezelfde  
  ↳ uitkomst hebben, se = FALSE, want R studio gaf een foutmelding.  
ggplot(movies, aes(x = log_budget, y = log_revenue)) + geom_point(alpha = 0.5, color =  
  ↳ "black") +  
geom_smooth(method = "lm", se = FALSE, color = "red") +  
labs( title = "Log Revenue vs Log Budget", x = "Log Budget", y = "Log Revenue") +  
theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 280 rows containing non-finite outside the scale range  
## (`stat_smooth()`).
```

```
## Warning: Removed 280 rows containing missing values or values outside the scale range  
## (`geom_point()`).
```



Your Answer:

The scatterplot is consistent with the previously found b_1 value of 0.885, it shows a clear positive correlation between log budget and log revenue. Furthermore showing a wide range of scatter around the average line, thus showing budget cannot fully predict revenue.

step b

```
#[] gebruik ik om iets te selecteren uit de dataset
movies_complete <- movies[complete.cases(movies$log_revenue, movies$log_budget), ]
movies_complete$predicted_log_revenue <- predict(model4.2, newdata = movies_complete)
movies_complete$residual <- movies_complete$log_revenue -
  ↪ movies_complete$predicted_log_revenue
top_movie <- movies_complete %>% slice_max(residual, n = 1)
top_movie[, c("title", "log_revenue", "predicted_log_revenue", "residual", "budget",
  ↪ "revenue")]
```

```
## # A tibble: 1 x 6
##   title      log_revenue predicted_log_revenue residual budget revenue
##   <chr>      <dbl>          <dbl>         <dbl>   <dbl>   <dbl>
## 1 Tarnation    14.0              7.33         6.63    218 1162014
```

Your Answer:

The movie that has vastly overperformed percentage wise the most is Tarnation, using 218 dollars to make 1,162,014 dollars. Tarnation has a log revenue of 13.97, a predicted log revenue of 7.33, and a residual of 6.63, the largest in the entire dataset.

2 Week 5

- Make a frequency table of the variable genre. Which are the three genres that occur the most? [2 points]
- Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called "Other". Report a frequency table of this new variable genre2. [2 points]
- Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. [4 points]

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,000. It will have a runtime of 120 minutes.

- Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. [5 points]
- Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? [2 points]
- One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? [5 points]

step a

```
# Frequentie tabel van de genre variabel
genre_freq_table <- table(movies$genre)
# Sorteert om de top 3 te vinden
sort(genre_freq_table, decreasing = TRUE)[1:3]
```

```
##
##      Drama      Comedy Thriller
##      455        195        127
```

Your Answer:

The three most occurring genres in the dataset are Drama, Comedy and Thriller.

step b

```
# Creëer een variabel die Drama, Comedy en Thriller laat zien en de rest bij elkaar voegt
  ↳ onder 'Other'
movies$genre2 <- ifelse(movies$genre %in% c("Drama", "Comedy", "Thriller"),
                       movies$genre,
                       "Other")
# Laat de tabel zien met de nieuwe variabele
table(movies$genre2)
```

```
##
##      Comedy      Drama      Other Thriller
##      195        455        132        127
```

Your Answer:

The new variable 'genre2' retains Drama, Comedy and Thriller, while compacting all remaining genres into the 'Other' category.

step c

```
# Alle 0en als NA's weergeven
movies$log_revenue <- ifelse(movies$revenue == 0, NA, log(movies$revenue))
movies$log_budget <- ifelse(movies$budget == 0, NA, log(movies$budget))

# Schat OLS model
model_c <- lm(log_revenue ~ log_budget + factor(genre2) + release_year, data = movies)
summary(model_c)
```

```
##
## Call:
## lm(formula = log_revenue ~ log_budget + factor(genre2) + release_year,
##     data = movies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.8744  -0.6599   0.1682   0.7996   6.4989
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -24.607849   18.022645  -1.365    0.173
## log_budget      0.863306    0.044720  19.305 <2e-16 ***
## factor(genre2)Drama    -0.096063    0.153515  -0.626    0.532
## factor(genre2)Other     0.234793    0.204194   1.150    0.251
## factor(genre2)Thriller   0.023238    0.191402   0.121    0.903
## release_year     0.013741    0.008977   1.531    0.126
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.476 on 623 degrees of freedom
## (280 observations deleted due to missingness)
## Multiple R-squared:  0.4079, Adjusted R-squared:  0.4032
## F-statistic: 85.85 on 5 and 623 DF, p-value: < 2.2e-16
```

Your Answer: Intercept(-24.6078): The intercept represents the expected log revenue of a baseline comedy movie released in the year 0, with a total production budget of \$1. Because a budget of \$1 and a release year of 0 are impossible, this coefficient has no meaningful economic interpretation.

Log_budget (0.8633): Because both budget and revenue are log-transformed, this represents the elasticity. A 1% increase in the production budget leads to an approximate 0.86% increase in revenue, whilst holding other variables constant.

factor(genre2): The coefficients represent the expected difference in log revenue compared to the baseline Comedy category. In a log-linear model, the percentage effect is calculated as $100 * (\exp(\text{coefficient}) - 1)$. For example, movies in the "Other" category have a coefficient of 0.2348, meaning approximately 26.5% higher expected revenue than comedies when holding budget and year constant. Similarly, Drama and Thriller have approximately 9.1% lower and 2.3% higher revenues relative to comedy, respectively

Release_year (0.0137): Each successive year of release is associated with an approximate 1.37% increase in revenue, holding all other factors constant.

step d

```
# Runtime in model zetten om specifieke film te schatten
model_d <- lm(log_revenue ~ log_budget + factor(genre2) + release_year + runtime, data =
  ↪ movies)

# Dataframe maken voor 2026 drama movie (met budget van 20,000,000)
new_movie <- data.frame(
  log_budget = log(20000000),
  genre2 = "Drama",
  release_year = 2026,
  runtime = 120
)

# Schat log omzet met 90% prediction interval
pred_log <- predict(model_d, newdata = new_movie, interval = "prediction", level = 0.90)

# Exponenteer om resultaten in USD te krijgen
exp(pred_log)
```

```
##          fit      lwr      upr
## 1 49301798 4262893 570191984
```

Your Answer:

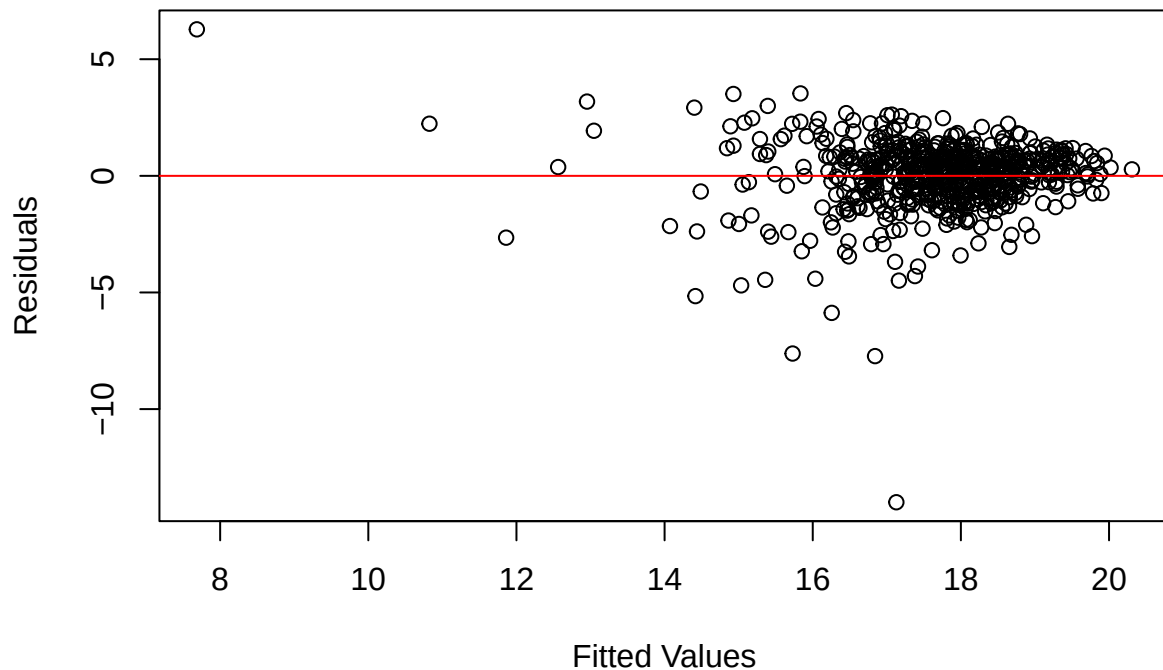
After building a model that accounts for runtime, the predicted revenue for this 2026 Drama movie is around \$49.3 million. The 90% prediction interval ranges from \$4.26 million to \$570.2 million, showing a high variance in movie revenues.

*Note: The question mentions a budget of “20,000,0000”. Course coordinator Aglio confirmed that this is supposed to be \$20 million.

step e

```
# Plot de residuals tegen fitted values om heteroskedasticity te checken
plot(model_d$fitted.values, resid(model_d),
  main = "Residuals vs. Fitted Values",
  xlab = "Fitted Values",
  ylab = "Residuals")
abline(h = 0, col = "red")
```

Residuals vs. Fitted Values



Your Answer:

By plotting the residuals against the fitted values (or other independent variables), we can visually inspect the spread. If the spread of the residuals widens or narrows significantly as fitted values increase (a funnel shape) this provides evidence of heteroskedasticity.

This plot shows clear evidence of heteroskedasticity because the vertical variance of the residuals is not constant. The errors are highly dispersed for fitted values between 14 and 17 but cluster more tightly as fitted values approach 20, perfectly matching the 'funnel shape'.

step f

```
# Schat een nieuw model met release_month als een categorisch variabele
model_f <- lm(log_revenue ~ log_budget + factor(genre2) + release_year + runtime +
  ↪ factor(release_month), data = movies)

# Voer een F-test (anova) uit die de twee modellen vergelijkt
anova(model_d, model_f)
```

```
## Analysis of Variance Table
##
## Model 1: log_revenue ~ log_budget + factor(genre2) + release_year + runtime
## Model 2: log_revenue ~ log_budget + factor(genre2) + release_year + runtime +
##   factor(release_month)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      622 1346.7
## 2      611 1320.1 11    26.626 1.1203 0.3422
```

Your Answer:

The F-test comparing the model with and without the `release_month` variable has a p-value of 0.3422. Because this p-value is greater than the standard 0.05 significance level, it proves that the release month does not have a statistical significant impact on a movie's revenue. Therefore, I would advise the studio that trying to time the release to a specific month to maximize revenue is not supported by this.